

2.1 Objects and Collections

In earlier sections, we introduced expressions and discussed the meanings they convey. Meaningful expressions refer to *objects*.

Notion 2.1 (Object). An **object** is anything to which a meaningful expression may refer.

For example, a personal name refers to a person, a numeral refers to a number, a quoted expression refers to an expression. All these are examples of objects.

Often, we must refer to several objects collectively rather than individually. To do so, we group objects into collections.

Notion 2.2 (Collection). A **collection** is a selection of objects, which are its **elements**.

To name collections, I will normally use expressions written starting bold with upper case letters.

Notation 2.3 (Names for collections). Expressions of the form $\mathbf{A}e$, where $\mathbf{A}$ is a bold upper-case letter and e is any expression containing letters or numerals will serve as names for collections.

Examples of names for collections following this pattern are:

$\mathbf{A}$, $\mathbf{B}e_1$, $\mathbf{Z}\mathbf{Z}_\mathbf{A}$, $\mathbf{A}\mathbf{B}$, $\mathbf{I}n^1$, $\mathbf{A}a$, $\mathbf{S}tuff$, $\mathbf{S}tufff$.

The following, instead, will not name collections:

$\mathbf{A}$, $\mathbf{B}e_1$, $\mathbf{Z}\mathbf{Z}_\mathbf{A}$, $\mathbf{A}\mathbf{B}$, $^1\mathbf{I}n$, $a\mathbf{A}$, $\mathbf{S}tufff$, $\mathbf{S}tuff$.

Historical note: The notion of a collection

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One of the exceptions to the convention above will be my use of the double-struck upper case letters ' $\mathbb{B}$ ', ' $\mathbb{N}$ ', ' $\mathbb{W}$ ', ' $\mathbb{Z}$ ', ' $\mathbb{Q}$ ', ' $\mathbb{R}$ ', and ' $\mathbb{C}$ ' to denote the sets of boolean, natural, whole, integer, rational, real, and complex numbers, respectively.

Collections come in two kinds: *sets* and *tuples*. In the following, I will provide intuitive and informal account of these notions to be used through this textbook.

2.2 Sets and members

A *set* is a collection in which we disregard any order or repetitions concerning its elements aka members.

Notion 2.4 (Set). A **set** is a collection in which the elements are neither ordered nor repeated.

The sign ' $\in$ ' stands for the *set membership* relation.

Notation 2.5 (Set membership). The expression ' $e \in S$ ', where e stands for an object and S stands for a set, is an abbreviation of ' e is a member of S '. Moreover, ' $e \notin S$ ' abbreviates ' $\neg e \in S$ '.

Thus, for example, ' $a \in A$ ' means that a is a member of A , and ' $a \notin A$ ' means that a is not a member of A .

Task 2.1. Abbreviate the following expressions using **notations 2.5 and 1.39**.

1. 1 is a natural number.
2. A is the set of natural numbers.
3. 1 is not 0.
4. 0 is not a natural number.
5. 1 is not the set of natural numbers.
6. Natural numbers and real numbers are not the same.
7. The letter ' a ' is in the set A .
8. The set of natural numbers is in the set of rational numbers.

Hint: A collection can be an element of another collection.

Note first that the expressions in this task are quite different from those used when the notation was introduced. They are all formulated by means of the verb *to be*, which has several meanings in natural language, and which is represented by means of different logical and set-theoretical notions depending on the context. This task is an opportunity to see these different uses.

Historical note: The notion of a set

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*: Both expressions are true under the usual convention for natural numbers. In particular, 0 is placed in the set of *whole numbers*, usually represented by ' $\mathbb{W}$ ' or ' $\mathbb{N}_0$ ', which consists of all natural numbers together with 0.

In Expressions 2.1.1 and 2.1.4, 'is' indicates that an object falls under a certain collection. Saying that 1 is a natural number, or that 0 is not a natural number means to place (or exclude) these numbers within a collection. Thus, this use of 'is' is rendered by the membership relation, and these expressions would be abbreviated as ' $1 \in \mathbb{N}$ ' and ' $0 \notin \mathbb{N}$ ', respectively.*

A different use appears in Expressions 2.1.2, 2.1.3, and 2.1.6. Here, 'is', 'is not', and 'are' express sameness or difference between objects. In these cases, the appropriate notion is identity, rather than membership. Hence, these expressions should be abbreviated as ' $\mathbf{A} = \mathbb{N}$ ', ' $1 \neq 0$ ' and ' $\mathbb{N} \neq \mathbb{R}$ ', respectively.

Expressions 2.1.5, 2.1.7, and 2.1.8 are trickier.

Rather than saying, like Expression 2.1.1, that 1 is not a natural number Expression 2.1.5 says 1 is not the (set of) natural numbers. That is, the number 1 and the natural numbers are not the same object. This expression is true, and should be translated as ' $1 \neq \mathbb{N}$ '.

Expression 2.1.7 is not saying that the object a is in the set $\mathbf{A}$. It rather says that the letter ' a ' – that is, a linguistic sign – is in the set $\mathbf{A}$. This requires us a way to distinguishing between a sign and its referent, which in this case we do through quote marks.

Finally, Expression 2.1.8 is quite ambiguous and may be understood in at least two ways.

First, it may be understood as stating that $\mathbb{N}$ is an element of $\mathbb{R}$, which should be represented with ' $\mathbb{N} \in \mathbb{R}$ '. This statement is false, since, although all the natural numbers are real numbers, the set of natural numbers is not itself a number and, hence, it cannot be a member of $\mathbb{R}$ nor of any other set of numbers.

Second, it may be understood as stating that $\mathbb{N}$ is a part of or is *included* in $\mathbb{R}$. This would be true if understood in the sense that all the natural numbers are real numbers. We cannot represent this

understanding with the notation introduced so far, but we will acquire the notation needed in [definition 2.11](#).

Solution to task 2.1.

- | | |
|--|---|
| 1. 1 is a natural number. | Abbreviation: $1 \in \mathbb{N}$ |
| 2. A is the set of natural numbers. | Abbr.: $\mathbf{A} = \mathbb{N}$ |
| 3. 1 is not 0. | Abbr.: $1 \neq 0$ |
| 4. 0 is not a natural number. | Abbr.: $0 \notin \mathbb{N}$ |
| 5. 1 is not the set of natural numbers. | Abbr.: $1 \neq \mathbb{N}$ |
| 6. Natural numbers and real numbers are not the same. | Abbr.: $\mathbb{N} \neq \mathbb{R}$ |
| 7. The letter ' <i>a</i> ' is in the set A . | Abbr.: ' <i>a</i> ' $\in \mathbf{A}$ |
| 8. The set of natural numbers is in the set of rational numbers. | Possible abbreviation: $\mathbb{N} \in \mathbb{Q}$ |

A notable set is the *empty set*, which has no elements.

Notion 2.6 (Empty set). The **empty set** is the set with no members.

Notation 2.7 (Empty set). ' $\{\}$ ' denotes the empty set.

Describing sets. There are two usual ways to describe the contents a set. For the purposes of this explanation, let **A** be a set containing the numbers 1, 2, 3, and nothing else as members.

The first way is by enumeration, that is, by listing all of its elements between braces.

Notation 2.8 (Set by enumeration). ' $\lceil \{e\} \rceil$ ', where *e* stands for an object, denotes the set having that object and nothing else as a member. Moreover, ' $\lceil \{1, e\} \rceil$ ', where *l* is a comma-separated list of expressions standing for objects and *e* stands for an object, denotes the set having all those objects and nothing else as members.

The order in which the elements are listed in this notation is irrelevant, and so is whether the elements are repeated.

For instance, the expression ' $\{1, 2, 3\}$ ' adequately describes **A**, which entails that $\mathbf{A} = \{1, 2, 3\}$. Hence:

$$\mathbf{A} = \{1, 2, 3\} = \{2, 1, 3\} = \{2, 1, 2, 3\} = \{3, 2, 1, 1, 2, 3\}.$$

This notation is suitable for describing small, finite sets, but not for large or infinite sets.

Historical note: Set-builder notation

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The second way is by extension, that is, by specifying a property common to all its members. For instance, **A** may be correctly described as the set of all natural numbers which are smaller than 4. To express this, we use the *set-builder notation*.

Notation 2.9 (Set-builder). The expression $\lceil \{t : A\} \rceil$, where t stands for an unspecified object and A is a propositional sentence, denotes the objects denoting by t which make A is true.

Thus, to describe **A** in the terms above, we write:

$$\mathbf{A} = \{n : n \in \mathbb{N} \ \& \ n < 4\}.$$

In words, **A** is equal to the set of all n such that n is a natural number ($n \in \mathbb{N}$) and n is smaller than 4 ($n < 4$).

A more elegant way to put it is as follows:

$$\mathbf{A} = \{n \in \mathbb{N} : n < 4\}.$$

In words, **A** is equal to the set of all natural numbers n ($n \in \mathbb{N}$) such that n is smaller than 4 ($n < 4$). In this case, the expression to the left of ‘:’ indicates the domain from which the elements of the set are drawn, while the expression to the right specifies the particular property that these elements must satisfy.

Not that an expression like ‘ $\mathbf{A} = \{1, 2, 3\}$ ’ is a *propositional sentence* stating that **A** is the set which contains the numbers 1, 2, 3, and nothing else, which may be true or false.

Sometimes, we do not want to make an assertion about the identity between two sets, but rather to provide a **definition** of set name in terms of our notation. In such cases, instead of the equality symbol ‘=’, we use the definition symbol ‘ $\stackrel{\text{def}}{=}$ ’.

Task 2.2. Determine the elements of the sets defined below.

- $$\begin{aligned}\mathbf{A} &\stackrel{\text{DF}}{=} \{x : x \text{ is or was a Pope}\}. \\ \mathbf{B} &\stackrel{\text{DF}}{=} \{x : x \text{ is a member of the Beatles}\}. \\ \mathbf{C} &\stackrel{\text{DF}}{=} \{x : x \text{ is the name of a member of the Beatles}\}. \\ \mathbf{D} &\stackrel{\text{DF}}{=} \{x : x \text{ is an uppercase vowel in English}\}. \\ \mathbf{E} &\stackrel{\text{DF}}{=} \{x \in \mathbb{N} : x \geq 10 \wedge x \leq 20\}. \\ \mathbf{F} &\stackrel{\text{DF}}{=} \{x \in \mathbb{R} : x \geq 10 \wedge x \leq 20\}. \\ \mathbf{G} &\stackrel{\text{DF}}{=} \{m + n : m, n \in \mathbb{N} \wedge m, n < 10 \wedge n = 2m\}. \\ \mathbf{H} &\stackrel{\text{DF}}{=} \{x : x \in \mathbf{B} \wedge x \text{ is alive in 2025}\}. \\ \mathbf{I} &\stackrel{\text{DF}}{=} \{x \in \mathbf{B} : x \text{ is a member of Wings}\}. \\ \mathbf{J} &\stackrel{\text{DF}}{=} \{x \in \mathbb{N} : x \leq 10 \wedge x \geq 20\}.\end{aligned}$$

We first have sets whose members are people.

A collects all those who are or have been Popes – figures such as Saint Peter, John Paul II, Benedict XVI, and Francis. **B** also collects people, but in this case to the four members of the Beatles: John Lennon, Paul McCartney, George Harrison, and Ringo Starr.

H and **I** are closely related to **B**. The former contains those Beatles who are still alive in 2025 (Paul McCartney and Ringo Starr), whereas the latter singles out the Beatle who was also part of the band Wings (Paul McCartney).

The next sets concern words and letters.

C contains the names of the Beatles – that is, the strings ‘John Lennon’, ‘Paul McCartney’, ‘George Harrison’, and ‘Ringo Starr’ – not John Lennon, Paul McCartney, George Harrison, and Ringo Starr themselves.

D contains the uppercase vowels of English: ‘A’, ‘E’, ‘I’, ‘O’, and ‘U’. However, A, E, I, O, and U are not members of **D**. This answer assumes that ‘A’, ‘E’, ‘I’, ‘O’, and ‘U’ are not used – as it occurs in ordinary written language – as names for ‘A’, ‘E’, ‘I’, ‘O’, and ‘U’. We do not want to do that in this textbook.

Next, we have some numerical sets.

E and **F** both include all numbers greater than or equal to 10 and less than or equal to 20, the former restricted to natural numbers and the latter extended to real numbers.

G consists of sums of two natural numbers below 10 such that one is twice the other; this gives the values: $1 + 2 = 3$, $2 + 4 = 6$, $3 + 6 = 9$, and $4 + 8 = 12$.

Finally, **J** has no members, since no natural number can be simultaneously less than 10 and greater than 20.

Historical note: The membership symbol

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Solution to task 2.2.

- ▶ Saint Peter, ..., Francis, Leo XIV $\in \mathbf{A}$.
- ▶ John Lennon, Paul McCartney, George Harrison, Ringo Starr $\in \mathbf{B}$.
- ▶ 'John Lennon', 'Paul McCartney', 'George Harrison', 'Ringo Starr' $\in \mathbf{C}$.
- ▶ 'A', 'E', 'I', 'O', 'U' $\in \mathbf{D}$.
- ▶ 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 $\in \mathbf{E}$.
- ▶ All members of $\mathbf{D}$ and all real numbers between them are members of $\mathbf{F}$.
- ▶ 3, 6, 9, 12 $\in \mathbf{G}$.
- ▶ Paul McCartney, Ringo Star $\in \mathbf{H}$.
- ▶ Paul McCartney $\in \mathbf{I}$.
- ▶ $\mathbf{J}$ has no members.

Paradoxical sets. Using the set builder notation does not commit us to regard as mathematically or philosophically acceptable to form a set from any statements.

To explain this point, remember that sets are objects too and, hence, can also be members of sets. Next, note that some member might be said to be members of themselves. Thus for instance, the set of mathematical entities is a mathematical entity, so it would be a member of itself. Other sets are clearly not members of themselves, like the set of living beings, which is no living being.

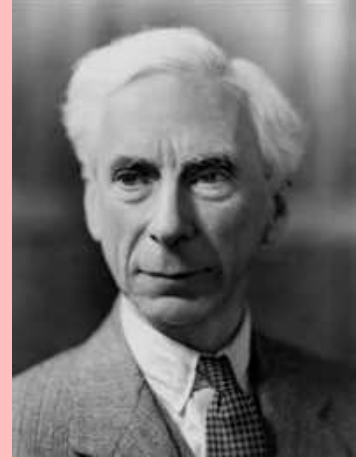
Now, consider the purported set:

$$\mathbf{R} = \{\mathbf{S} : \mathbf{S} \notin \mathbf{S}\}$$

Historical note: Russell's paradox

Though first noticed by Ernst Zermelo, Russell's paradox was first published by Bertrand Russell in his *Principles of Mathematics* (1903, §100). Here is how he explains it in a later work:

[N]ormally a class is not a member of itself. Mankind, for example, is not a man. Form now the assemblage of all classes which are not members of themselves. This is a class: is it a member of itself or not? If it is, it is one of those classes that are not members of themselves, i.e. it is not a member of itself. If it is not, it is not one of those classes that are not members of themselves, i.e. it is a member of itself. Thus of the two hypotheses—that it is, and that it is not, a member of itself—each implies its contradictory. (1919, p. 136)



B. Russell (1872–1970)

In other words, $\mathbf{R}$ is the set of all sets which are not members of themselves.

Now assume that $\mathbf{R} \in \mathbf{R}$. Consequently, $\mathbf{R}$ is a member of itself, which means that $\mathbf{R} \notin \mathbf{R}$. Since this contradicts with our assumption $\mathbf{R} \in \mathbf{R}$, we must conclude that $\mathbf{R} \notin \mathbf{R}$. However, this means that $\mathbf{R}$ is not a member of itself, which entails that $\mathbf{R} \in \mathbf{R}$.

We can then see that, at least in this reasoning, the existence of $\mathbf{R}$ is paradoxical, as it entails a contradiction. Thus, we should probably consider that no set $\mathbf{R}$, as described above, exists.

This paradox is known as *Russell's paradox*.

The lesson to take from this, at least for now, is that the set-builder notation must not be taken as license to make-up set in terms of a given expression at the left of ':'. It just allows us to speak about a set, if it exist, through this notation.

2.3 Operations and Relations on Sets

Union, intersection, difference. Given some sets, we can apply various operations to them in order to describe associated sets. The following defines the more standard such operations on sets:

Definition 2.10 (Set operations).

$$A \cup B \stackrel{\text{DF}}{=} \{x : x \in A \text{ } \mathbb{W} \text{ } x \in B\} \quad (\text{Union})$$

$$A \cap B \stackrel{\text{DF}}{=} \{x : x \in A \text{ } \mathbb{M} \text{ } x \in B\} \quad (\text{Intersection})$$

$$A \setminus B \stackrel{\text{DF}}{=} \{x : x \in A \text{ } \mathbb{M} \text{ } x \notin B\} \quad (\text{Difference})$$

: A, B stand for sets.

*: In some notations, we use ' $\subset$ ' for the *subset* relation and ' $\subsetneq$ ' for the *proper subset* relation (see below).

Subset, superset. Now, sets may be related in certain ways. Importantly, when a set contains all the members of another set, the latter is a *subset* of the former, and the former is a *superset* of the latter.*

Definition 2.11 (Subset and superset).

$$A \subseteq B \stackrel{\text{DF}}{=} \forall x (x \in A \Rightarrow x \in B). \quad (\text{Subset})$$

$$A \supseteq B \stackrel{\text{DF}}{=} B \subseteq A. \quad (\text{Superset})$$

: A, B stand for sets.

Extensionality. An important intuitive principle about sets is that they are defined solely by their *extension*, that is, by the elements they contain. Hence, if any two sets have the same elements, they are actually one and the same set.

†: More typically, we would express this axiom as follows:

$$\forall x (x \in A \Leftrightarrow x \in B) \Leftrightarrow A = B$$

: A, B stand for sets.

This principle is known as the *axiom of extensionality*, which we might express (a bit unorthodoxly) with the following axiom schema[†]:

Axiom 2.1 (Extensionality).

$$A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A : A, B \text{ stand for sets.}$$

‡: The notation ' $\cap$ ' is idiosyncratic.

More relations between sets. Other important relations include the following[‡]:

Definition 2.12 (Set relations).

$$A \subset B \stackrel{\text{DF}}{=} A \subseteq B \wedge A \neq B \quad (\text{Proper subset})$$

$$A \supset B \stackrel{\text{DF}}{=} B \subset A \quad (\text{Proper superset})$$

$$A \cap B \stackrel{\text{DF}}{=} A \cap B = \{\} \quad (\text{Disjointness})$$

: A, B stand for sets.

Power set. As we have seen, set can also contain other sets as members. For example:

$$\{\{1, 2\}, \{a, b\}, \{3, 4\}, \{'a', 'za'\}\}.$$

A whole kind of sets are constructed via the **power set** operation. The power set of A , written ' $\wp A$ ' or ' 2^A ', is the set of all subsets of A .

Definition 2.13 (Power set).

$$\wp A \stackrel{\text{DF}}{=} \{X : X \subseteq A\}$$

: A stands for a set, X is an unspecified set.

Historical note: Naïve set theory

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Author (XXXX-XXXX)

Note that the empty set, $\{\}$, is a subset of any set and, hence, an element of the power set of any set.

Corollary 2.14. For all sets A , $\{\} \subseteq A$.

Proof. Let A be an arbitrarily chosen set. Note that $\{\}$ has no elements. Hence, all $x \in \{\}$ are such that $x \in A$ (and $x \notin \{\}$, and whatever we want). This, by [definition 2.12](#), entails that $\{\} \subseteq A$. Since A was arbitrarily chosen, this holds for all sets A . $\square$

Corollary 2.15. For all sets A , $\{\} \in \wp A$.

Proof. Let A be an arbitrarily chosen set. By [Corollary 2.14](#), we know that $\{\} \subseteq A$, which, by [definition 2.13](#), entails that $\{\} \in \wp A$. Since A was arbitrarily chosen, this holds for all sets A . $\square$

Task 2.3. Consider the following sets:

$$\begin{aligned} A &= \{n \in \mathbb{N} : 0 \leq n < 4\}, & B &= \{n \in \mathbb{N} : 1 < n \leq 4\}, \\ C &= A \cap B, & D &= A \cup B, \\ E &= B \setminus A, & F &= C \cup E. \end{aligned}$$

- For each pair A, B of these sets, determine whether: (i) $A \subseteq B$, (ii) $A = B$, (iii) $A \subset B$, (iv) $A \not\subset B$.
- Now consider their respective power sets $\wp A, \wp B, \wp C, \wp D, \wp E, \wp F$. Do these relations still hold for these sets too?

First, note that, in general, for all sets A , we have: $A \subseteq A$, $A = A$, not $A \subset A$, and not $A \not\subset A$. As for the relationships between sets with

different names, it is be useful to determine their exact contents:

$$\begin{aligned} \mathbf{A} &= \{n \in \mathbb{N} : 0 \leq n < 4\} = \{1, 2, 3\}, \\ \mathbf{B} &= \{n \in \mathbb{N} : 1 < n \leq 4\} = \{2, 3, 4\}, \\ \mathbf{C} &= \mathbf{A} \cap \mathbf{B} = \{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}, \\ \mathbf{D} &= \mathbf{A} \cup \mathbf{B} = \{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}, \\ \mathbf{E} &= \mathbf{B} \setminus \mathbf{A} = \{1, 2, 3\} \setminus \{2, 3, 4\} = \{4\}, \\ \mathbf{F} &= \mathbf{C} \cup \mathbf{E} = \{2, 3\} \cup \{4\} = \{2, 3, 4\} = \mathbf{B}, \end{aligned}$$

We can now easily determine the relations between **A**, **B**, **C**, **D**, **E**, **F**.

Regarding their power sets, things are a bit different. In particular, note that, whereas $\mathbf{A} \pitchfork \mathbf{E}$, this does not hold for their power sets. By [Corollary 2.14](#), the empty set $\{\}$ is an element – not just a subset – that $\wp \mathbf{A}$ and $\wp \mathbf{E}$ have in common, that is, $\wp \mathbf{A} \cap \wp \mathbf{E} = \{\{\}\}$.

Solution to task 2.3.

1. The tables below show how the sets are related. For example, in the $\subseteq$ -table, each cell at row **A** and column **B** contains ‘yes’ if $\mathbf{A} \subseteq \mathbf{B}$, and ‘no’ otherwise.

$\subseteq$	A	B	C	D	E	F
A	yes	no	no	yes	no	no
B	no	yes	no	yes	no	yes
C	yes	yes	yes	yes	no	yes
D	no	no	no	yes	no	no
E	no	yes	no	yes	yes	yes
F	no	yes	no	yes	no	yes

$=$	A	B	C	D	E	F
A	yes	no	no	no	no	no
B	no	yes	no	no	no	yes
C	no	no	yes	no	no	no
D	no	no	no	yes	no	no
E	no	no	no	no	yes	no
F	no	yes	no	no	no	yes

$\subset$	A	B	C	D	E	F
A	no	no	no	yes	no	no
B	no	no	no	yes	no	no
C	yes	yes	no	yes	no	yes
D	no	no	no	no	no	no
E	no	yes	no	yes	no	yes
F	no	no	no	yes	no	no

$\pitchfork$	A	B	C	D	E	F
A	no	no	no	no	yes	no
B	no	no	no	no	no	no
C	no	no	no	no	no	no
D	no	no	no	no	no	no
E	yes	no	no	no	no	no
F	no	no	no	no	no	no

2. Most of these relationships remain exactly the same for the power sets. However, this does not hold for the disjointness relationship, because, for all sets **A**, **B**, we have $\{\} \in \wp \mathbf{A} \cap \wp \mathbf{B}$.

2.4 Numerals and Numbers

Digits and Numerals. In this section, we want to define a special kind of expressions: *numerals*. These are names for numbers, e.g., ‘0’, ‘2’, ‘55’, ‘100’, etc. Note, however, that ‘00’, ‘01’, ‘000989’ are not usually regarded as numerals.

A way to do so is by defining the term ‘is a numeral’ via a clause-based inductive definition to be predicated of all expressions having the form of numerals.

In order to fix our base cases, we will first explicitly define the term 'digit' to cover numerals consisting of exactly one character.

$$\begin{aligned} d \text{ is a digit} &\stackrel{\text{DF}}{=} d = '0' \vee d = '1' \vee d = '2' \vee d = '3' \vee d = '4' \vee \\ &\quad d = '5' \vee d = '6' \vee d = '7' \vee d = '8' \vee d = '9' \quad (\Upsilon) \end{aligned}$$

The definition must make sure that, in the inductive clause, the first number is not a numeral.

Task 2.4. Provide a clause-based inductive definition of 'numeral'.

Hint: Note that all digits are numerals, but no multi-digit numeral starts with '0'.

Solution to task 2.4.

Definition 8. The term 'numeral' is recursively defined by the following clauses:

1. $d \text{ is a digit} \Rightarrow d \text{ is a numeral.}$ *(base clause)*
2. $n \text{ is a numeral} \wedge n \neq '0' \wedge d \text{ is a digit} \Rightarrow nd \text{ is a numeral.}$ *(inductive clause)*
3. No e is a numeral unless it can be obtained by finitely many applications of 1 and 2. *(final clause)*

Note that the following definition of a numeral would be incorrect:

Mistake 2.16. The term 'numeral' is recursively defined by the following clauses:

1. $d \text{ is a digit} \Rightarrow d \text{ is a numeral.}$ *(base clause)*
2. $d \text{ is a digit} \wedge d \neq '0' \wedge n \text{ is a numeral} \Rightarrow \lceil dn \rceil \text{ is a numeral.}$ *(inductive clause)*
3. No e is a numeral unless it can be obtained by finitely many applications of 1 and 2. *(final clause)*

Task 2.5. Explain why the definition in **mistake 2.16** is wrong.

The definition is intended to generate exactly the set of numerals, that is, strings of digits that either consist of the single digit '0' or begin with a non-zero digit followed by arbitrary digits. However, the given recursive definition fails to generate all such strings.

The error lies in the direction of concatenation used in the inductive clause. According to the base clause, every digit is a numeral. Hence, the numerals initially obtained are:

'0', '1', '2', '3', '4', '5', '6', '7', '8', '9'.

Historical note: The positional system

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The inductive clause then states that if d is a digit different from '0' and n is a numeral, then dn is a numeral. Thus, new digits are always added to the *left* of an already existing numeral, and the added digit must be non-zero.

Because of this, it is impossible to append digits to the right of an existing numeral. In particular, numerals consisting of more than one digit and ending in '0' cannot in general be generated.

For example, consider the numeral '100'. From the base clause we obtain '0' and '1'. Using the inductive clause, we can form '10' from '1' and '0'. However, there is no way to obtain '100', because this would require appending another '0' to the right of '10', which is not permitted by the inductive clause. The clause only allows placing a non-zero digit on the left of an already constructed numeral.

More generally, once the rightmost digit of a numeral has been fixed by the base clause, it can never be changed. Hence, many valid numerals, such as '100', '250', or '40', cannot be obtained by the given rules.

Solution to task 2.5.

The definition in **mistake 2.16** is incorrect because its inductive clause only allows adding digits to the left of a numeral. Thus digits cannot be appended to the right. For example, '10' can be formed, but '100' cannot, since this would require adding a '0' to the right of '10'. Hence, not all numerals can be generated.

The clause-based definitions above can become somewhat lengthy. A more convenient way to define numerals is via defining sets of digits and then sets of numerals.

We will define two sets of digits: **Lea**, the set of *leading digits*, which includes all digits from '1' to '9', and **Dig**, the set of *digits*,

which also includes '0'. Similarly, we define two sets of numerals: $\mathbf{Num}_{\mathbb{N}}$, the set of *natural numerals*, which includes all numerals starting from '1', and $\mathbf{Num}_{\mathbb{W}}$, the set of *whole numerals*, which also includes '0'.

Task 2.6. Define the sets $\mathbf{Lea}$, $\mathbf{Dig}$, $\mathbf{Num}_{\mathbb{N}}$ and $\mathbf{Num}_{\mathbb{W}}$.

Hint: Take inspiration from definition (I), on page 41.

Definition 2.17 (solution to task 2.6).

$$\mathbf{Lea} \stackrel{\text{DF}}{=} \{'1', '2', '3', '4', '5', '6', '7', '8', '9'\}$$

$$\mathbf{Dig} \stackrel{\text{DF}}{=} \{'0'\} \cup \mathbf{Lea}$$

$$\mathbf{Num}_{\mathbb{N}} \stackrel{\text{DF}}{=} \mathbf{Lea} \cup \{nd : n \in \mathbf{Num}_{\mathbb{N}} \wedge d \in \mathbf{Dig}\}$$

$$\mathbf{Num}_{\mathbb{W}} \stackrel{\text{DF}}{=} \{'0'\} \cup \mathbf{Num}_{\mathbb{N}}$$

Note that, despite appearances, **definition 2.17** is inductive, as its definiens states both the base case (to the left of 'U') and its inductive case (to its right).

Zero and successor. Having introduced numerals as expressions, we now turn to *number*, the objects that numerals denote. For example, the numeral '0' denotes the number zero, '2' denotes the number two, '55' denotes the number fifty-five, etc. Thus, numerals are linguistic objects, whereas numbers are abstract objects.

A basic theory of numbers can be expressed with a reduced language containing just the numeral '0' and the successor operator 'S(n)', where n stands for a number.

Intuitively, '0' represents the null quantity; for instance, to say that there are 0 dogs is to say that there are no dogs. The successor operator 'S' denotes the operation of going to the next number. For example, the successor of 0 is the number one, the successor of one is two, and so on.

The fundamental properties of numbers are captured by the following axioms, known as the *Peano axioms*.

Axiom 2.2 (Peano axioms). Where 'n' is a number variable and K is an arbitrary set, the set $\mathbb{W}$ of whole numbers is such that:

1. $0 \in \mathbb{W}$.
2. $\forall n (n \in \mathbb{W} \Rightarrow S(n) \in \mathbb{W})$.
3. $\neg \exists n (n \in \mathbb{W} \wedge S(n) = 0)$.
4. $\forall m \forall n (S(m) = S(n) \Rightarrow m = n)$.
5. $(0 \in K \wedge \forall n (n \in K \Rightarrow S(n) \in K)) \Rightarrow \mathbb{W} \subseteq K$. (Induction)

Historical note: Peano Axioms

G. Peano (1858–1932)

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Axiom 2.2 characterises all the properties of whole numbers. In particular, we can now introduce the basic arithmetic operations. The following are some important theorems of Peano axioms.

Theorem 2.18. $\forall n (n \neq 0 \Rightarrow \exists m (n = S(m)))$.

Theorem 2.19. $\forall n (n \neq S(n))$.

The *sum* operation, written $\ulcorner m + n \urcorner$ (where m, n stand for numbers) combines two numbers into a new number. Intuitively, $(m + n)$ is the result of adding n units to m . We can define addition recursively.

Definition 2.20 (Sum).

$$\begin{aligned} (m + 0) &\stackrel{\text{DF}}{=} m \\ (m + S(n)) &\stackrel{\text{DF}}{=} S(m + n) \\ &: m, n \text{ stand for numbers.} \end{aligned}$$

The operation of *multiplication*, written $\ulcorner m \cdot n \urcorner$ represents repeated addition. Intuitively, $(m \cdot n)$ is the result of adding m to itself n times. Like sum, multiplication is defined recursively.

Definition 2.21 (Multiplication).

$$\begin{aligned} (m \cdot 0) &\stackrel{\text{DF}}{=} 0 \\ (m \cdot S(n)) &\stackrel{\text{DF}}{=} (m \cdot n) + m \\ &: m, n \text{ stand for numbers.} \end{aligned}$$

We can now easily introduce the usual numerals.

Definition 2.22 (Numerals).

$$\begin{aligned}
 1 &\stackrel{\text{DF}}{=} S(0), & 2 &\stackrel{\text{DF}}{=} S(1), & 3 &\stackrel{\text{DF}}{=} S(2), \\
 4 &\stackrel{\text{DF}}{=} S(3), & 5 &\stackrel{\text{DF}}{=} S(4), & 6 &\stackrel{\text{DF}}{=} S(5), \\
 7 &\stackrel{\text{DF}}{=} S(6), & 8 &\stackrel{\text{DF}}{=} S(7), & 9 &\stackrel{\text{DF}}{=} S(8), \\
 10 &\stackrel{\text{DF}}{=} S(9), & m1 &\stackrel{\text{DF}}{=} (m \cdot 10) + 1, & m2 &\stackrel{\text{DF}}{=} (m \cdot 10) + 2, \\
 m3 &\stackrel{\text{DF}}{=} (m \cdot 10) + 3, & m4 &\stackrel{\text{DF}}{=} (m \cdot 10) + 4, & m5 &\stackrel{\text{DF}}{=} (m \cdot 10) + 5, \\
 m6 &\stackrel{\text{DF}}{=} (m \cdot 10) + 6, & m7 &\stackrel{\text{DF}}{=} (m \cdot 10) + 7, & m8 &\stackrel{\text{DF}}{=} (m \cdot 10) + 8, \\
 m9 &\stackrel{\text{DF}}{=} (m \cdot 10) + 9, & n0 &\stackrel{\text{DF}}{=} (n \cdot 10) \\
 & & & & : m \in \mathbf{Num}_{\mathbb{W}} \setminus \{0\} \ \& \ n \in \mathbf{Num}_{\mathbb{W}} \setminus \{0, 1\}.
 \end{aligned}$$

Below are some important consequences of these definitions.

Theorem 2.23. $\forall n (0 + n = n)$.

Theorem 2.24. $\forall m \forall n (m + n = n + m)$.

Theorem 2.25. $\forall l \forall m \forall n ((l + m) + n = l + (m + n))$.

Theorem 2.26. $\forall n (0 \cdot n = 0)$.

Theorem 2.27. $\forall n (1 \cdot n = n)$.

Theorem 2.28. $\forall m \forall n (m \cdot n = n \cdot m)$.

Theorem 2.29. $\forall l \forall m \forall n ((l \cdot m) \cdot n = l \cdot (m \cdot n))$.

Theorem 2.30. $\forall l \forall m \forall n (l \cdot (m + n) = (l \cdot m) + (l \cdot n))$.

Strict and non-strict order. In addition to arithmetic operations, numbers admit an *order* relationship expressing when one number is less than another.

In particular, we have a relator $\ulcorner m < n \urcorner$, which reads ' m is less than n ', expressing the relation of *strict order*.

Definition 2.31 (Strict order for numbers).

$$\begin{aligned}
 (m < n) &\stackrel{\text{DF}}{=} \exists x (x \in \mathbb{N} \ \& \ (m + x) = n) \\
 & : m, n \text{ stand for numbers.}
 \end{aligned}$$

Intuitively, $(m < n)$ holds precisely when n can be obtained from m by adding at least 1. That is, there is at least one step of separation

Historical note: Order

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from m to n in the succession chain.

We may also define a relator $\ulcorner m \leq n \urcorner$, which reads ' m is less than or equal to n ', expressing the relation of *non-strict order*. Moreover, we have the converses $\ulcorner m \geq n \urcorner$ and $\ulcorner m > n \urcorner$ of the relations above, which read ' m is greater than or equal to n ' and ' m is greater n ', respectively.

Definition 2.32 (Order).

$$(m > n) \stackrel{\text{DF}}{=} (n < m)$$

$$(m \leq n) \stackrel{\text{DF}}{=} (m < n) \mathbin{\wedge} (m = n)$$

$$(m \geq n) \stackrel{\text{DF}}{=} (n \leq m)$$

Intuitively, $(m \leq n)$ holds precisely when n can be obtained from m by adding some number, which may be 0. Thus, it holds between m and m itself, and between m and all its successors.

The following are consequences of the definition of order.

Theorem 2.33. $\forall n (0 \leq n)$

Theorem 2.34. $\forall n (n \leq n)$.

Theorem 2.35. $\forall m \forall n (m \leq n \mathbin{\wedge} n \leq m \Rightarrow m = n)$.

Theorem 2.36. $\forall l \forall m \forall n (l \leq m \mathbin{\wedge} m \leq n \Rightarrow l \leq n)$.

Theorem 2.37. $\forall m \forall n (m < n \mathbin{\vee} m = n \mathbin{\vee} n < m)$.

Theorem 2.38. $\forall k \forall m \forall n (m \leq n \Rightarrow m + k \leq n + k)$.

Theorem 2.39. $\forall k \forall m \forall n (m < n \Rightarrow m + k < n + k)$.

Theorem 2.40. $\forall k \forall m \forall n (m \leq n \Rightarrow m \cdot k \leq n \cdot k)$.

Theorem 2.41. $\forall k \forall m \forall n (m < n \wedge k \neq 0 \Rightarrow m \cdot k < n \cdot k)$.

Theorem 2.42. $\forall m \forall n (m < n \Rightarrow S(m) \leq n)$.

Axiom 2.2 together with **definitions 2.20, 2.21** and **2.32** provide the basic structure needed to carry out arithmetic within the system of whole numbers.

Order and numerals. Note that **definition 2.31** applies to numbers, but not between numerals. It is useful to extend this relation to numerals too, so that this relation holds between two numerals provided it holds between its denoted numbers.

However, also note that **definition 2.31** is only defined for numbers, while **definition 2.32** has no restriction whatsoever. This means that, if we extend the relation ' $>$ ' to numerals, the **definition 2.32** would apply to them. We can do so by defining an operation analogous to ' S ' for numerals: the next numeral operation ' N '.

Definition 2.43 (Next).

$$\begin{aligned} N(em) = en &\stackrel{\text{DF}}{=} (m = '0' \wedge n = '1') \vee (m = '1' \wedge n = '2') \vee \\ &\quad (m = '2' \wedge n = '3') \vee (m = '3' \wedge n = '4') \vee \\ &\quad (m = '4' \wedge n = '5') \vee (m = '5' \wedge n = '6') \vee \\ &\quad (m = '6' \wedge n = '7') \vee (m = '7' \wedge n = '8') \vee \\ &\quad (m = '8' \wedge n = '9') \\ N(fm) = en &\stackrel{\text{DF}}{=} N(f) = e \wedge m = '9' \wedge n = '0' \end{aligned}$$

On this basis, we extend the relation ' $>$ ' to numerals as follows:

Definition 2.44 (Strict order for numerals).

$$(m < n) \stackrel{\text{DF}}{=} \exists x (x \in \mathbb{N} \wedge (m + x) = n)$$

: m, n stand for numbers.

This results in the following desired theorem:

Theorem 2.45. Let m, n be numerals and m, n , respectively, the numbers they stand for: $m < n \Leftrightarrow m < n$.

Historical note: Cardinality

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Note that **definitions 2.43** and **2.44** not only apply numerals, but also to any expression which ends with a numeral. This will include numerals occurring as subindexes, which we will introduce in the next section.

The operation ‘N’ and the relation ‘<’ can also be extended to letters, so that ‘ a ’ < ‘ b ’, ‘ b ’ < ‘ c ’, and so forth, also for upper-case letters. In fact, we can extended to any expressions among which we can find some intuitive ordering.

Cardinality. The *cardinality* of a set expresses the idea of how many elements the set contains. It provides a way to compare sets in terms of their size, whether they are finite or infinite.

A set **A** is said to be finite iff its elements can be listed one by one and the listing ends. The cardinality of a set is simply the number of distinct elements it has.

We write ‘ $|A|$ ’ to denote the cardinality of **A**. For instance:

$$|\{1, 2, 3\}| = 3, \quad |\{a, b\}| = 2, \quad |\{\}| = 0.$$

Two finite sets have the same cardinality if there exists a one-to-one correspondence (a **bijection**) between their elements.

The following definition express all this:

Definition 2.46 (Cardinality).

$$|\{\}| \stackrel{\text{DF}}{=} 0$$

$$|\{a\}| \stackrel{\text{DF}}{=} 1$$

$$|A \cup B| \stackrel{\text{DF}}{=} |A| + |B| - |A \cap B|$$

: **A**, **B** stand for sets $\mathbb{M}$ **a** stands for an object.

A set is *infinite* iff it is not finite, that is, iff its elements cannot be fully listed in a finite sequence. Even so, infinite sets can also be compared by their cardinalities.

A classical example is the set of natural numbers:

$$\mathbb{N} = \{0, 1, 2, 3, \dots\}.$$

Its cardinality is denoted by the symbol ' $\aleph_0$ ' (read 'aleph-null'), which is the smallest infinite cardinality.

Definition 2.47 (Cardinality of $\mathbb{N}$). $\aleph_0 \stackrel{\text{DF}}{=} |\mathbb{N}|$.

Although we cannot go into the proof, we will state the following important facts:

Fact 2.48. $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| = \aleph_0$.

Fact 2.49. $|\mathbb{R}| > \aleph_0$.

2.5 Sub-indexes and Ellipsis

Sub-indexed placeholders. Our list of basic placeholders provided in [Section 1.4](#) seems to be sufficient to express anything we want to say about unspecified expressions. We have seventy-four in total, and I am sure I will not use more than ten placeholders in any schema used or mentioned in this textbook.

However, this does not mean that the sixty-six basic placeholders will suffice for saying all we want to say. Sometimes, we will want to speak not just about one, two, or seventy-four unspecified expressions, but about an unspecified number of them; for instance, n such unspecified expressions.

To do so, we need to be able to have infinitely many placeholders. Actually, at least one for each whole number. A way to achieve this is to take all placeholders introduced in [notation 1.16](#) and create new placeholders constructed from their concatenation with subindexed numerals.

Let us improve our definition of a placeholder. First, we define a set **BEPla** containing our basic expression placeholders, and then we define the set **SEPla** of subindexed basic placeholders. We next merge these two sets into **EPla**, containing all our expression placeholders.

Definition 2.50 (Placeholder).

$$\mathbf{BEPla} \stackrel{\text{DF}}{=} \{b : b \text{ is a basic placeholder}\}.$$

$$\mathbf{SEPla} \stackrel{\text{DF}}{=} \{b_n : b \in \mathbf{BEPla} \ \& \ n \in \mathbf{Num}_{\mathbb{W}}\}.$$

$$\mathbf{EPla} \stackrel{\text{DF}}{=} \mathbf{BEPla} \cup \mathbf{SEPla}.$$

What **definition 2.50** says is that any expression of the form b_n is a subindexed placeholder. Note that the placeholder 'n' in **definition 2.50** stands for a numeral. Thus, the 'n' to the right of 'b' and to the left of ':' denotes our numeral n but written in a smaller font size at the bottom, i.e., a subindex.

Complex placeholders. Subindexed placeholders are not always enough, or at least we can use a further extension. Sometimes, we need to write a placeholder to make explicit that some expression occurs within it. Focus in the statement of the following notation, as we will use meta-placeholders.

*: Thus, 'P' functions here as a meta-placeholder.

Notation 2.51. Let P be a placeholder symbol* and l a comma-separated list of expressions. The expression $\lceil P[l] \rceil$ is a placeholder for an expression P in which the expressions contained in the list l may occur (in no particular order). Moreover, where m is a comma-separated list of expressions with the same number of commas as l, $\lceil P[m] \rceil$ denotes the expression resulting from replacing P each element of l with a corresponding element of m.

†: Except, of course, in quotations.

Ellipsis. The reader might have noticed that a symbol has not appeared in this textbook so far† which would have been quite used and useful in logic textbooks: the ellipsis, that is, '...'.
 This is an informal symbol used in mathematics, and which is not normally defined neither rigorously nor informally. The following defines '...' as it will be understood throughout this textbook (outside quotations).

Definition 2.52 (Ellipsis). The expression '...' is recursively defined as follows:

$$E[N(n)] \blacksquare \dots \blacksquare E[m] \stackrel{\text{DF}}{=}$$

$$E[m] \blacksquare \dots \blacksquare E[m] \stackrel{\text{DF}}{=} E[m]$$

$$E[m] \blacksquare \dots \blacksquare E[N(n)] \stackrel{\text{DF}}{=} E[m] \blacksquare \dots \blacksquare E[n] \blacksquare E[N(n)]$$

$$E[m] \blacksquare \dots \stackrel{\text{DF}}{=} E[m] \blacksquare \dots \blacksquare E[u]$$

$$: m, n, u \in \mathbf{Num}_{\mathbb{W}} \ \& \ m \leq n \ \& \ u \geq m.$$

In this definition, ' $\blacksquare$ ' is a placeholder for a separating expression, like a comma. However, nothing prevents from applying this

Historical note: Ellipsis

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notation in an expression of the form:

$$\lceil S(1) + \dots + S(n) \rceil,$$

where '+' takes the place of '■'.

Also note that, in the third clause of **definition 2.52**, we the first expression is of the form $\lceil E[N(n)] \rceil$, where $N(n)$ is strictly less than m , the numeral occurring in the last expression, i.e., in $E[m]$. In this case, this denotes the empty expression, and not a sequence in the reverse direction.

Finally, note that the last clause **definition 2.52** results in expressions of the form $\lceil E[1] \dots \rceil$ being always schemata. Hence, expressing them functions as expressing each of instance of $\lceil E[1] \dots \rceil E[u] \rceil$.

Now, the reason we did not define '...' before is that it was a bit difficult to state without numerals and an order relation defined for them. It is now possible to re-state some of our previous notions and notations more clearly.

For instance, if we assume the obvious ordering of letters, we may use ellipsis to express the following fact about placeholders.

Fact 2.53 (Placeholder). The type-writer font letters and subindexed letters:

'A', 'A₀', 'A₁', ... 'B', 'B₀', 'B₁', ... 'Y', 'Y₀', 'Y₁', ... 'Z', 'Z₀', 'Z₁', ...
'a', 'a₀', 'a₁', ... 'b', 'b₀', 'b₁', ... 'y', 'y₀', 'y₁', ... 'z', 'z₀', 'z₁', ...

as well as:

'■', '■ ₀ ', '■ ₁ ', ...,	'□', '□ ₀ ', '□ ₁ ', ...,
'●', '● ₀ ', '● ₁ ', ...,	'○', '○ ₀ ', '○ ₁ ', ...,
'⦿', '⦿ ₀ ', '⦿ ₁ ', ...,	'⊙', '⊙ ₀ ', '⊙ ₁ ', ...,
'♥', '♥ ₀ ', '♥ ₁ ', ...,	'♡', '♡ ₀ ', '♡ ₁ ', ...,
'♣', '♣ ₀ ', '♣ ₁ ', ...,	'♠', '♠ ₀ ', '♠ ₁ ', ...,

$\langle \mathbf{1}, \mathbf{1}_0, \mathbf{1}_1, \dots \rangle, \langle \mathbf{2}, \mathbf{2}_0, \mathbf{2}_1, \dots \rangle, \dots$
 $\langle \mathbf{3}, \mathbf{3}_0, \mathbf{3}_1, \dots \rangle, \langle \mathbf{4}, \mathbf{4}_0, \mathbf{4}_1, \dots \rangle, \dots$
 $\langle \mathbf{5}, \mathbf{5}_0, \mathbf{5}_1, \dots \rangle, \langle \mathbf{6}, \mathbf{6}_0, \mathbf{6}_1, \dots \rangle, \dots$
 $\langle \mathbf{7}, \mathbf{7}_0, \mathbf{7}_1, \dots \rangle, \langle \mathbf{8}, \mathbf{8}_0, \mathbf{8}_1, \dots \rangle, \dots$
 $\langle \mathbf{9}, \mathbf{9}_0, \mathbf{9}_1, \dots \rangle, \langle \mathbf{0}, \mathbf{0}_0, \mathbf{0}_1, \dots \rangle, \dots$
 $\langle \mathbf{1}, \mathbf{1}_0, \mathbf{1}_1, \dots \rangle, \langle \mathbf{2}, \mathbf{2}_0, \mathbf{2}_1, \dots \rangle, \dots$

serve as **expression placeholders**.

Task 2.7. Re-state **notations 1.26** and **2.51** using subindexed placeholders and ellipses.

2.6 Tuples and Relations

Tuples. We have seen *sets*, which are collections of objects where order and repetition is not important. Collections which record the order and repetition of its objects are **tuples**.

Thus, a 2-tuple, aka **ordered pair** or **couple**, $\langle a_1, a_2 \rangle$ consists of two elements, and it is not the case in general that: $\langle a_1, a_2 \rangle = \langle a_2, a_1 \rangle$ or $\langle a_1, a_2 \rangle = \langle a_1, a_2, a_2 \rangle$. A 3-tuple, aka **ordered triple** or **triplet**, $\langle a, b, c \rangle$ consists of three elements. Let us state this notion in general.

Notion 2.54 (Tuple). A **tuple** is a collection in which the elements are ordered and can be repeated.

We may refer to an element of a tuple in a specific position via a *projection*.

Notation 2.55 (Projection). $\mathbf{T}[n]$ denotes the n -th member of $\mathbf{T}$: $\mathbf{T}$ is a tuple $n \in \mathbf{Num}_0 \wedge n \in \mathbb{N}$ and n stands for n .

This allows us to define a tuple via a projection. For instance:

$$\mathbf{TEv}[n] \stackrel{\text{DF}}{=} 2 \times n$$

Thus, $\mathbf{TEv}[1] = 2$, $\mathbf{TEv}[2] = 4$, $\mathbf{TEv}[3] = 6$, $\mathbf{TEv}[4] = 8$, etc.

A practical way to denote tuples with finitely many members is by enumeration.

Notation 2.56 (Tuple by enumeration).

$$\begin{aligned}
 \langle a_1, \dots, a_n \rangle [1] &\stackrel{\text{DF}}{=} a_1, \dots, \langle a_1, \dots, a_n \rangle [n] \stackrel{\text{DF}}{=} a_n \\
 &: n \in \mathbf{Num}_{\mathbb{N}}.
 \end{aligned}$$

Historical note: Cartesian products

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Cartesian product. Given two sets **A** and **B**, the **Cartesian product** of **A** and **B**, written '**A** × **B**', is the set of all ordered pairs with first component in **A** and second in **B**.

$$\mathbf{A} \times \mathbf{B} \stackrel{\text{DF}}{=} \{\langle a, b \rangle : a \in \mathbf{A} \ \& \ b \in \mathbf{B}\}.$$

Given sets $\mathbf{A}_1, \dots, \mathbf{A}_n$, their n -fold Cartesian product, written ' $\mathbf{A}_1 \times \dots \times \mathbf{A}_n$ ', is defined as follows:

Definition 2.57 (Cartesian product).

$$\begin{aligned} \mathbf{A}_1 \times \dots \times \mathbf{A}_n &\stackrel{\text{DF}}{=} \{\langle a_1, \dots, a_n \rangle : a_1 \in \mathbf{A}_1 \ \& \ \dots \ \& \ a_n \in \mathbf{A}_n\} \\ &: n \in \mathbf{Num}_{\mathbb{N}} \ \& \ n \text{ stands for } n \ \& \\ &\mathbf{A}_1, \dots, \mathbf{A}_n \text{ stand for sets} \ \& \ a_1, \dots, a_n \text{ are objects.} \end{aligned}$$

When we take the Cartesian product with respect to a single set, we use the following notation.

Notation 2.58.

$$\begin{aligned} \mathbf{A}^1 &\stackrel{\text{DF}}{=} \mathbf{A} \\ \mathbf{A}^{N(n)} &\stackrel{\text{DF}}{=} \mathbf{A}^n \times \mathbf{A} \\ &: \mathbf{A} \text{ stands for a set.} \end{aligned}$$

Relations. Cartesian products allow us to describe how elements of sets are associated with one another. Such associations are called *relations*. We will consider the case of relations of arity two.

Definition 2.59 (Two-place relation).

R is a relation from A to $B \stackrel{\text{DF}}{=} R \subseteq A \times B$
 $: A, B$ stand for sets.

In other words, a relation is a set of ordered pairs, which would contain the objects which stand in this relation to each other, in a given direction. Thus, if $\langle a, b \rangle \in R$, we say that ' a stands in the relation R with respect to b '.

Some basic notions associated with relations are the following.

Definition 2.60 (Domain and range of a relation).

$\text{Dom}(R) \stackrel{\text{DF}}{=} \{x \in A : \exists y (y \in B \wedge \langle x, y \rangle \in R)\}$
 $\text{Ran}(R) \stackrel{\text{DF}}{=} \{y \in B : \exists x (x \in A \wedge \langle x, y \rangle \in R)\}$
 $: R \subseteq A \times B.$

Thus, for $R \subseteq A \times B$, we have that $\text{Dom}(R)$ is the set of all elements of A which stand in the relation R to some element of B , in that order; and $\text{Ran}(R)$ is the set of all elements of B which stand in the relation R to some element of A , but in the reverse order.

Definition 2.61 (Image and preimage).

$\text{Pre}_R(X) \stackrel{\text{DF}}{=} \{x \in X : \exists y (y \in B \wedge \langle x, y \rangle \in R)\}$
 $\text{Ima}_R(Y) \stackrel{\text{DF}}{=} \{y \in Y : \exists x (x \in A \wedge \langle x, y \rangle \in R)\}$
 $: R \subseteq A \times B \wedge X \subseteq A \wedge Y \subseteq B.$

Thus, where $R \subseteq A \times B$, $X \subseteq A$, and $Y \subseteq B$, we have that $\text{Pre}_R(X)$ is the set of all elements of X which stand in the relation R to some element of B , in that order; and $\text{Ima}_R(Y)$ is the set of all elements of Y which stand in the relation R to some element of A , but in the reverse order.

Functions. A *function* may be regarded as a special kind of relation in which each element of the domain is associated with exactly one element of the codomain. Let us consider one-place functions.

Definition 2.62 (Function).

$$\begin{aligned} f : A \rightarrow B &\stackrel{\text{DF}}{=} f \subseteq A \times B \wedge \\ &\forall x (x \in A \wedge \exists! y (y \in B \wedge \langle x, y \rangle \in f)) \\ &: A, B \text{ stand for sets.} \end{aligned}$$

The expression $\ulcorner f : A \rightarrow B \urcorner$ reads ' f is a function from A to B '. We have moreover the following notation.

Notation 2.63 (Function).

$$\begin{aligned} f(a) = b &\stackrel{\text{DF}}{=} \langle a, b \rangle \in f \\ &: a, b \text{ stand for objects.} \end{aligned}$$